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Blockchain Experiment 3

Aim : Create a Cryptocurrency using Python and perform mining in the Blockchain created.

Theory :

1. **Blockchain Overview**

A blockchain is a **distributed, decentralized digital ledger** used to record transactions securely across multiple computers (nodes). Instead of relying on a central authority, every node in the network maintains its own copy of the blockchain.
Each block in the blockchain contains:

- A list of transactions
- A timestamp
- The hash of the previous block
- Its own cryptographic hash

The blocks are linked together using hashes, forming a chain. Any attempt to modify data in a block changes its hash, which breaks the chain and makes tampering easily detectable. This property ensures **data integrity, transparency, and immutability**.

2. **Mining**

Mining is the process by which new blocks are added to the blockchain. It involves:

- Collecting pending transactions
- Creating a block header
- Solving a computationally difficult problem known as **Proof-of-Work (PoW)**

In Proof-of-Work, miners repeatedly change a nonce value to generate a hash that satisfies the network's difficulty requirement (such as leading zeroes). Once a valid hash is found, the block is added to the blockchain and shared with other nodes. As an incentive, miners receive a **cryptocurrency reward** for their computational effort.

3. Multi-Node Blockchain Network

In this experiment, a **multi-node blockchain network** is simulated using three nodes running on different ports (5001, 5002, and 5003).
Each node:

- Operates independently
- Maintains its own copy of the blockchain
- Communicates with peer nodes to share newly mined blocks

This setup demonstrates how blockchain achieves decentralization and synchronization without a central server.

4. Consensus Mechanism

To ensure consistency across all nodes, the **Longest Chain Rule** is used as the consensus mechanism.
When different versions of the blockchain exist, the node accepts the **longest valid chain** as the correct one. This rule ensures that all nodes eventually agree on a single transaction history, even if temporary differences arise.

5. Transactions and Mining Reward

Transactions in the cryptocurrency system include:

- Sender address
- Receiver address
- Transaction amount

When a block is mined, all pending transactions are added to the block. Additionally, a special transaction is created to reward the miner with newly generated cryptocurrency. This reward mechanism motivates miners to maintain and secure the network.

6. Chain Replacement

The chain replacement process ensures network consistency. When the /replace_chain endpoint is triggered:

- A node requests blockchain data from its peers
- Validates the received chains
- Replaces its own chain if a longer and valid chain is found

This mechanism helps resolve conflicts and keeps all nodes synchronized.

Code :

```
import datetime

import hashlib

import json

from flask import Flask,jsonify,request
```

```
import requests

from uuid import uuid4

from urllib.parse import urlparse

class Blockchain:

    def __init__(self):

        self.chain=[]

        self.transactions=[]

        self.create_block(proof=1,previous_hash='0')

        self.nodes=set()

    def create_block(self,proof,previous_hash):

block={'index':len(self.chain)+1,'timestamp':str(datetime.datetime.now()),'proof':proof,'previous
_hash':previous_has h,'transactions':self.transactions}

        self.transactions=[]

        self.chain.append(block)

        return block

    def get_previous_block(self):

        return self.chain[-1]

    def proof_of_work(self,previous_proof):

        new_proof=1


        check_proof=False

        while not check_proof:
```

```
        hash_operation=hashlib.sha256(str(new_proof**2-previous_proof**2).encod
        e()).hexdigest() if hash_operation[:4]=='0000':
            check_proof=True
        else:
            new_proof+=1
    return new_proof
def hash(self,block):
    encoded_block=json.dumps(block,sort_keys=True).encode()
    return hashlib.sha256(encoded_block).hexdigest()
def is_chain_valid(self,chain):
    previous_block=chain[0]
    block_index=1
    while block_index<len(chain):
        block=chain[block_index]
        if block['previous_hash']!=self.hash(previous_block):
            return False
        previous_proof=previous_block['proof']
        proof=block['proof']
        hash_operation=hashlib.sha256(str(proof**2-previous_proof**2).encode
        ()).hexdigest() if hash_operation[:4]!='0000':
            return False
```

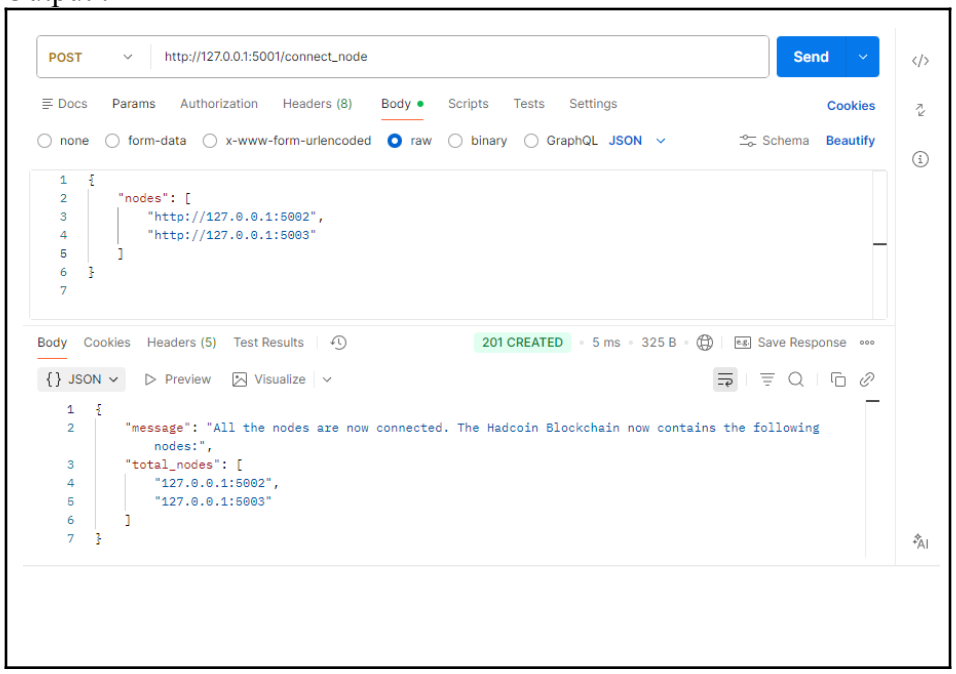
```
        previous_block=block
=

        block_index+=1

    return True

  def add_transaction(self,sender,receiver,amount):

    self.transactions.append({'sender':sender,'receiver':receiver,'amo

    unt':amount}) previous_block=self.get_previous_block()

    return previous_block['index']+1

  def add_node(self,address):

    parsed_url=urlparse(address)

    self.nodes.add(parsed_url.netloc)

  def replace_chain(self):

    network=self.nodes

    longest_chain=None

    max_length=len(self.chain)

    for node in network:

        response=requests.get(f'http://{node}/get_chain')

        if response.status_code==200:

            length=response.json()['length']

            chain=response.json()['chain']

            if length>max_length and self.is_chain_valid(chain):

                max_length=length
```

```
            longest_chain=chain
        if longest_chain:
            self.chain=longest_chain
            return True

        return False
app=Flask(__name__)
node_address=str(uuid4()).replace('-','')
blockchain=Blockchain()
@app.route('/mine_block',methods=['GET'])
def mine_block():
    previous_block=blockchain.get_previous_block()
    previous_proof=previous_block['proof']
    proof=blockchain.proof_of_work(previous_proof)
    previous_hash=blockchain.hash(previous_block)
    blockchain.add_transaction(sender=node_address,receiver='Richard',amount=1)
    block=blockchain.create_block(proof,previous_hash)
response={'message':'Congratulations, you just mined a
block!','index':block['index'],'timestamp':block['timestamp'],'proof':block['proof'],'previous_hash'
:block['previous_ha sh'],'transactions':block['transactions']}
    return jsonify(response),200
@app.route('/add_transaction',methods=['POST'])
def add_transaction():
```

```
  json_data=request.get_json()

  transaction_keys=['sender','receiver','amount']

  if not all(key in json_data for key in transaction_keys):

     return 'Some elements of the transaction are missing',400

  index=blockchain.add_transaction(json_data['sender'],json_data['receiver'],json

  _data['amount']) response={'message':f'This transaction will be added to Block

  {index}'}

  return jsonify(response),201
=

@app.route('/connect_node',methods=['POST'])

def connect_node():

  json_data=request.get_json()

  nodes=json_data.get('nodes')

  if nodes is None:

     return "No node",400

  for node in nodes:

     blockchain.add_node(node)

  response={'message':'All the nodes are now

  connected.','total_nodes':list(blockchain.nodes)} return jsonify(response),201

@app.route('/replace_chain',methods=['GET'])

def replace_chain():

  is_chain_replaced=blockchain.replace_chain()
```

```
    if is_chain_replaced:
        response={'message':'The chain was replaced by the longest
    one.','new_chain':blockchain.chain} else:
        response={'message':'The chain is already the largest
    one.','actual_chain':blockchain.chain} return jsonify(response),200
app.run(host='0.0.0.0',port=5000)
```

Output :



POST http://127.0.0.1:5002/connect_node

Send

Docs Params Authorization Headers (8) Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
{
    "nodes": [
        "http://127.0.0.1:5001",
        "http://127.0.0.1:5003"
    ]
}
```

Body Cookies Headers (5) Test Results 201 CREATED 8 ms 325 B Save Response

JSON Preview Visualize

```
{
    "message": "All the nodes are now connected. The Hadcoin Blockchain now contains the following nodes:",
    "total_nodes": [
        "127.0.0.1:5001",
        "127.0.0.1:5003"
    ]
}
```

New Collection / **New Request** Save Share

POST http://127.0.0.1:5003/connect_node

Send

Docs Params Authorization Headers (8) Body Scripts Tests Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
{
    "nodes": [
        "http://127.0.0.1:5001",
        "http://127.0.0.1:5002"
    ]
}
```

Body Cookies Headers (5) Test Results 201 CREATED 5 ms 325 B Save Response

JSON Preview Visualize

```
{
    "message": "All the nodes are now connected. The Hadcoin Blockchain now contains the following nodes:",
    "total_nodes": [
        "127.0.0.1:5002",
        "127.0.0.1:5001"
    ]
}
```

New Collection / **New Request**

Save | Share

POST http://127.0.0.1:5001/add_transaction **Send**

Docs | Params | Authorization | Headers (8) | Body | Scripts | Tests | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
{
    "sender": "Riya",
    "receiver": "Varyani",
    "amount": 22
}
```

Body | Cookies | Headers (5) | Test Results | 201 CREATED | 5 ms | 226 B | Save Response

JSON | Preview | Visualize

```
{
    "message": "This transaction will be added to Block 2"
}
```

New Collection / **New Request**

Save | Share

GET http://127.0.0.1:5001/mine_block **Send**

Docs | Params | Authorization | Headers (6) | Body | Scripts | Tests | Settings | Cookies

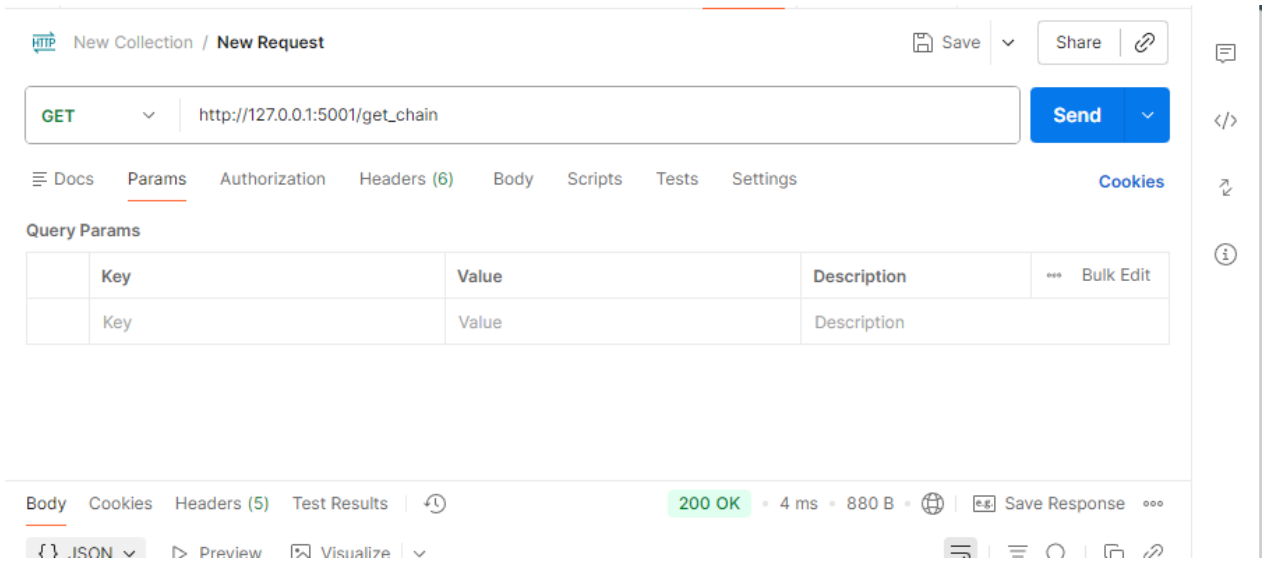
Query Params

	Key	Value	Description	Bulk Edit
	Key	Value	Description	

Body | Cookies | Headers (5) | Test Results | 200 OK | 4 ms | 513 B | Save Response

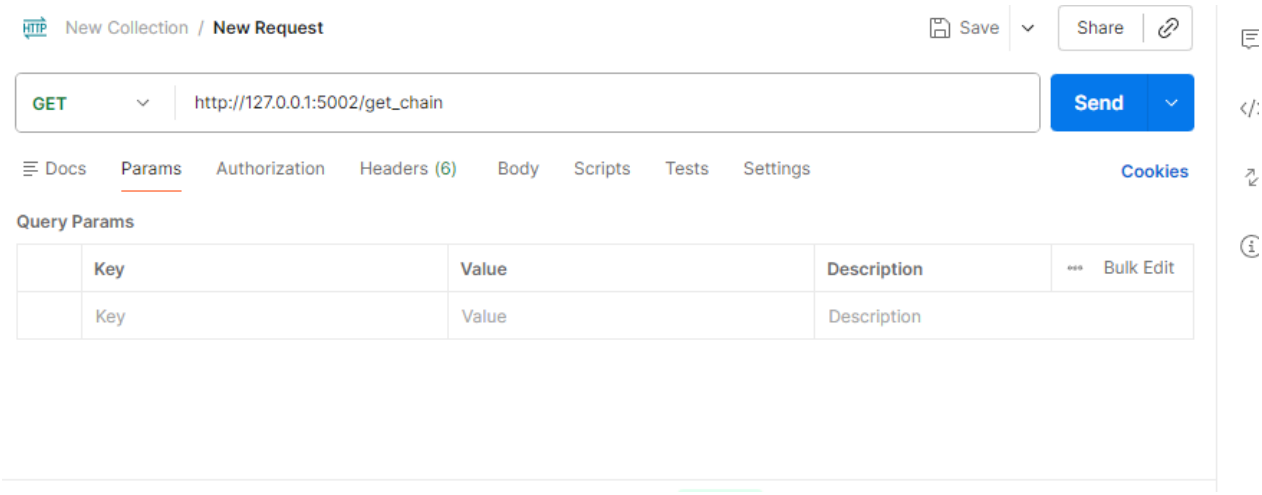
JSON | Preview | Visualize


```
{
    "index": 2,
    "message": "Congratulations, you just mined a block!",
    "previous_hash": "23baaa589072f85985d5ac34b0f9b44fae335f30794800fa7c7c49f952d147fe",
    "proof": 533,
    "timestamp": "2026-02-06 13:25:41.946952",
    "transactions": [
        {
            "amount": 22
```



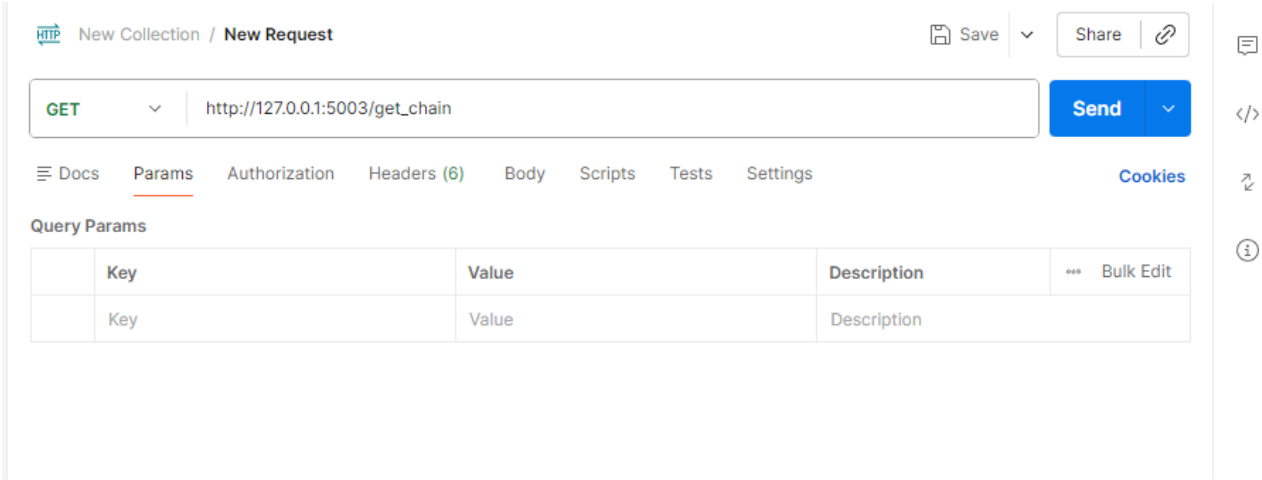

```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-06 12:58:40.478764",
            "transactions": []
        },
        {
            "index": 2,
                                                                "previous_hash":
"23baaa589072f85985d5ac34b0f9b44fae335f30794800fa7c7c49f952d147fe",
            "proof": 533,
            "timestamp": "2026-02-06 13:25:41.946952",
            "transactions": [
                {
                    "amount": 22,
                    "receiver": "Varyani",
                    "sender": "Riya"
                },
                {
                    "amount": 1,
                    "receiver": "Richard",
                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
            ]
        },
```

```
        {
            "index": 3,
                                                                "previous_hash":
"fcc5e4ebc37bb887e3a4e84cdd40e74bb0c3a473a7cc0a45b3c090c2cfd388e9",
            "proof": 45293,
            "timestamp": "2026-02-06 13:35:55.810244",
            "transactions": [
                {
                    "amount": 25,
                    "receiver": "Varyani",
                    "sender": "Neha"
                },
                {
                    "amount": 1,
                    "receiver": "Richard",
                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
            ]
        }
    ],
    "length": 3
}
```


```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
```

```
            "timestamp": "2026-02-06 12:58:40.478764",
            "transactions": []
        },
        {
            "index": 2,
                                                                "previous_hash":
"23baaa589072f85985d5ac34b0f9b44fae335f30794800fa7c7c49f952d147fe",
            "proof": 533,
            "timestamp": "2026-02-06 13:25:41.946952",
            "transactions": [
                {
                    "amount": 22,
                    "receiver": "Varyani",
                    "sender": "Riya"
                },
                {
                    "amount": 1,
                    "receiver": "Richard",
                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
            ]
        }
    ],
    "length": 2
}
```


```
{
    "chain": [
        {
            "index": 1,
```

```
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-06 12:58:40.478764",
            "transactions": []
        },
        {
            "index": 2,
                                                                "previous_hash":
"23baaa589072f85985d5ac34b0f9b44fae335f30794800fa7c7c49f952d147fe",
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                    "amount": 1,
                    "receiver": "Richard",
                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
            ]
        }
    ],
    "length": 2
}
```


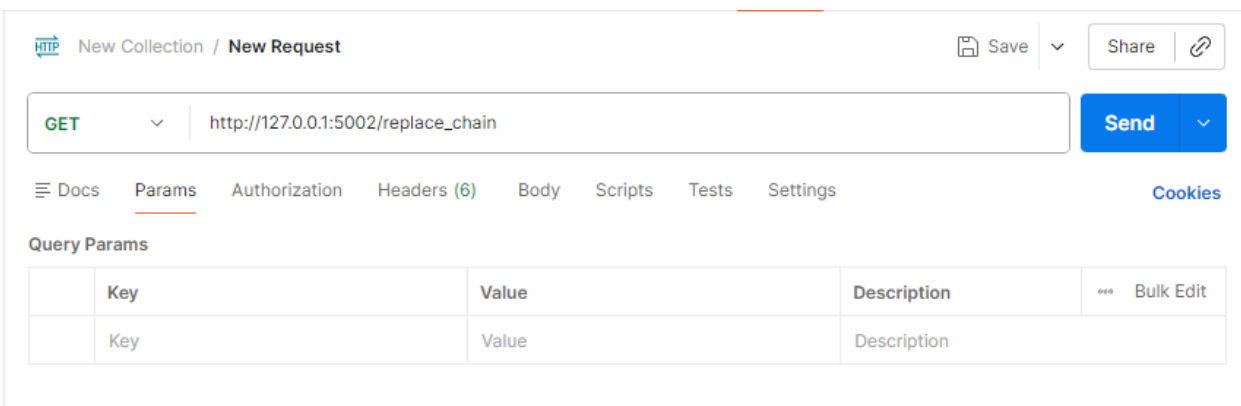


```
{
    "message": "The nodes had different chains so the chain was replaced by the longest one.",
    "new_chain": [
```

```
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-06 12:58:40.478764",
            "transactions": []
        },
        {
            "index": 2,
                                                                "previous_hash":
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                    "receiver": "Richard",
                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
            ]
        },
        {
            "index": 3,
                                                                "previous_hash":
"fcc5e4ebc37bb887e3a4e84cdd40e74bb0c3a473a7cc0a45b3c090c2cfd388e9",
            "proof": 45293,
            "timestamp": "2026-02-06 13:35:55.810244",
            "transactions": [
                {
                    "amount": 25,
                    "receiver": "Varyani",
                    "sender": "Neha"
                },
                {
                    "amount": 1,
```
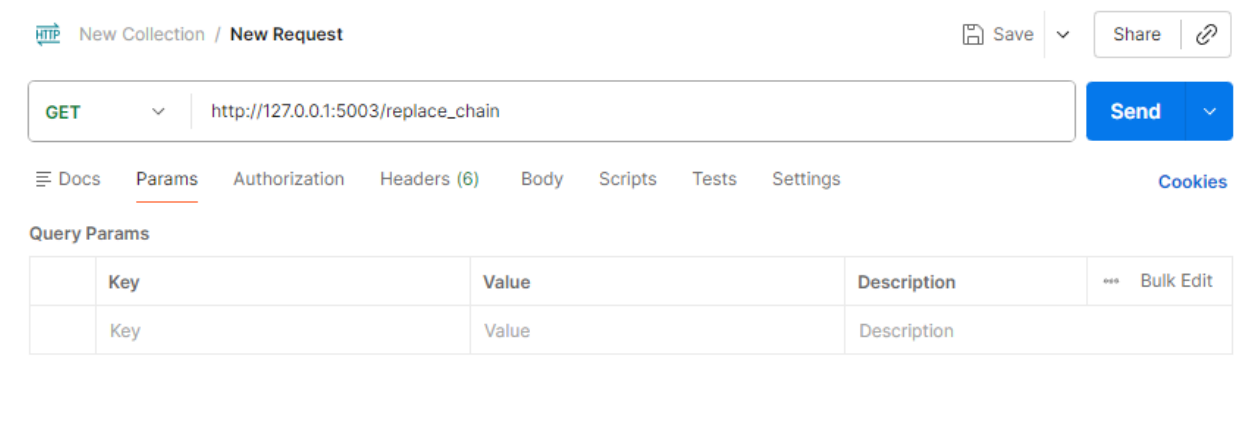

```
                "receiver": "Richard",
                "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
            }
        ]
    }
  ]
}
```


```
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    "message": "The nodes had different chains so the chain was replaced by the longest one.",
    "new_chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-06 12:58:40.478764",
            "transactions": []
        },
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                    "receiver": "Varyani",
                    "sender": "Riya"
                },
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                    "receiver": "Richard",
                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
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        {
            "index": 3,
                                                                "previous_hash":
"fcc5e4ebc37bb887e3a4e84cdd40e74bb0c3a473a7cc0a45b3c090c2cfd388e9",
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                    "amount": 1,
                    "receiver": "Richard",
                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
            ]
        }
    ]
}
```


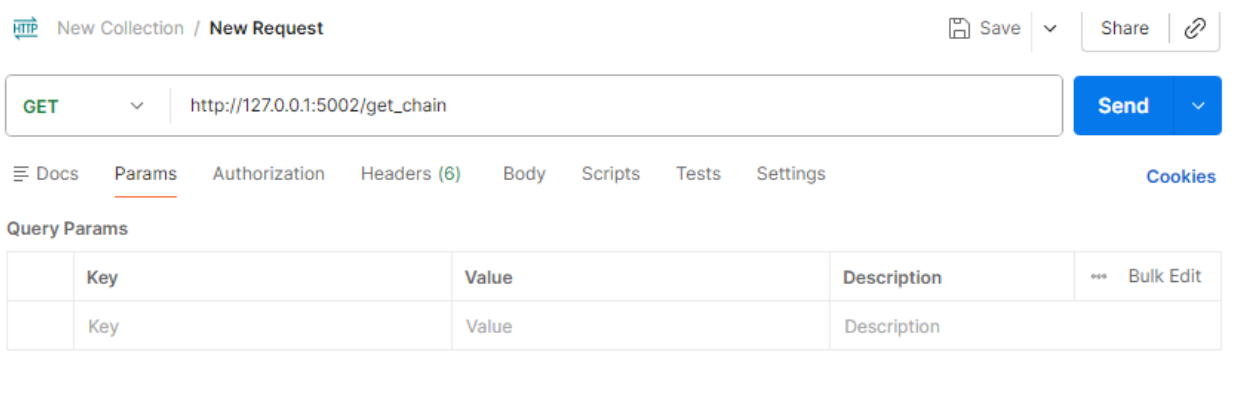


```
{
    "chain": [
        {
```

```
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-06 12:58:40.478764",
            "transactions": []
        },
        {
            "index": 2,
                                                                "previous_hash":
"23baaa589072f85985d5ac34b0f9b44fae335f30794800fa7c7c49f952d147fe",
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                {
                    "amount": 1,
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                    "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
                }
            ]
        },
        {
            "index": 3,
                                                                "previous_hash":
"fcc5e4ebc37bb887e3a4e84cdd40e74bb0c3a473a7cc0a45b3c090c2cfd388e9",
            "proof": 45293,
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                    "amount": 25,
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                    "sender": "Neha"
                },
                {
                    "amount": 1,
                    "receiver": "Richard",
```

```
                "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
            }
        ]
    }
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"length": 3
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```


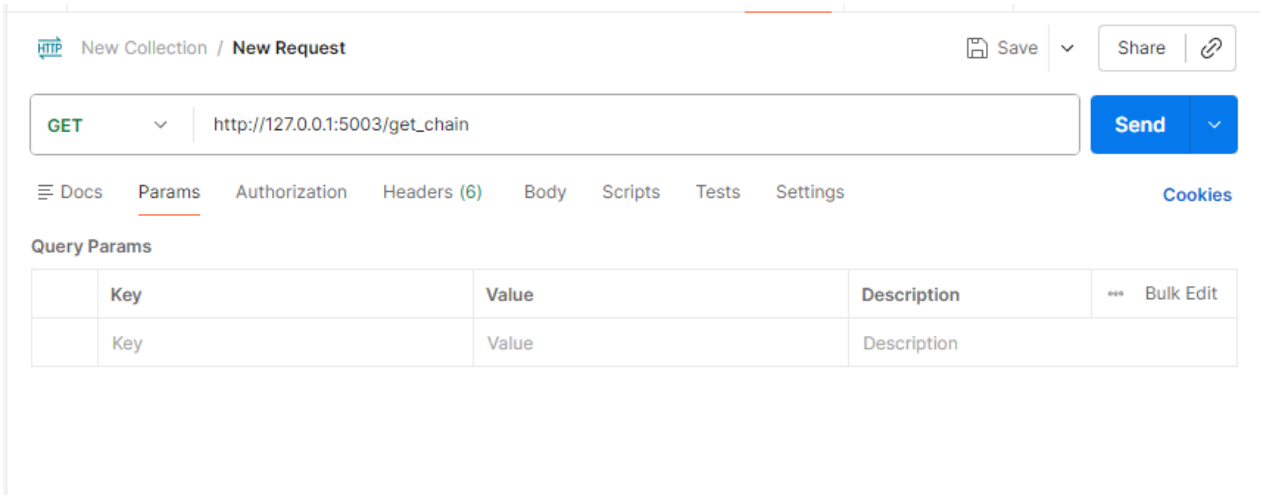


```
{
    "chain": [
        {
            "index": 1,
            "previous_hash": "0",
            "proof": 1,
            "timestamp": "2026-02-06 12:58:40.478764",
            "transactions": []
        },
        {
            "index": 2,
                                                                "previous_hash":
"23baaa589072f85985d5ac34b0f9b44fae335f30794800fa7c7c49f952d147fe",
            "proof": 533,
            "timestamp": "2026-02-06 13:25:41.946952",
            "transactions": [
                {
                    "amount": 22,
                    "receiver": "Varyani",
                    "sender": "Riya"
                },
                {
```

```
                "amount": 1,
                "receiver": "Richard",
                "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
            }
        ]
    },
    {
        "index": 3,
                                                                "previous_hash":
"fcc5e4ebc37bb887e3a4e84cdd40e74bb0c3a473a7cc0a45b3c090c2cfd388e9",
        "proof": 45293,
        "timestamp": "2026-02-06 13:35:55.810244",
        "transactions": [
            {
                "amount": 25,
                "receiver": "Varyani",
                "sender": "Neha"
            },
            {
                "amount": 1,
                "receiver": "Richard",
                "sender": "fc6c57a7d9f04e15aa9f10bf9e9f96d0"
            }
        ]
    }
  ],
  "length": 3
}
```


Conclusion : In this experiment, a simple cryptocurrency was successfully created using Python by implementing core blockchain concepts such as block creation, hashing, Proof-of-Work mining, transactions, and decentralization. The blockchain system was developed using Flask, allowing multiple nodes to communicate and maintain their own copies of the blockchain.

Mining was performed by solving the Proof-of-Work algorithm, through which new blocks were added to the blockchain and miners were rewarded with cryptocurrency. The experiment also demonstrated transaction handling, where transactions were verified and included in mined blocks. Overall, this experiment provided a practical understanding of how blockchain technology works, including mining, decentralization, and consensus, and highlighted how cryptocurrencies operate in real-world distributed systems.